

Measure-Theoretic Notions of Conditional Expectation

The statement that we will “condition on Y ,” implies that one should behave as if the value of Y is “known.”

$$E(X | Y = y)$$

We need to formalize this notion beyond conditioning on a single random variable, as this is crucial to more advanced study of probability. We will consider conditioning on more general forms of “information.”

It is useful to consider the following question: Given the nature of the information to be provided, for which subsets of the sample space Ω are we certain of their occurrence/non-occurrence?

The following example will help to build some intuition.

- Exercise:** Suppose that I will toss a coin three times. Let X denote the number of "heads." If I tell you that the information I will provide is the value of X , for which subsets of Ω are you certain of their occurrence/non-occurrence, regardless of the value of X ?

ω	$X(\omega)$	
HHH	3	$A_1 = \{HHH\}$
HHT	2	$A_2 = \{HHT, HTH, THH\}$
HTH	2	$A_3 = \{HTT, THT, TTH\}$
THH	2	$A_4 = \{TTT\}$
HTT	1	
THT	1	
TTH	1	
TTT	0	

For each of A_1, A_2, A_3, A_4 ,
know occ/non-occ based
on the value of X

But, will also know occ/non-occ of
 $\emptyset, A_1 \cup A_2, A_1 \cup A_3, \dots, A_1 \cup A_2 \cup A_3, \dots, \Omega$

would not know the occ/non-occ necessarily
 of $B = \{HHT\}$. If I tell you $X=2$,
 unsure if B has occurred or not.

Denote this collection of sets of

1 The above example demonstrates how “information” is captured via a col-
2 lection of subsets of Ω ; we will generally denote such collections using $\mathcal{F}$,
3 $\mathcal{G}$ or similar.

4 We will write $E(X|\mathcal{F})$ to indicate that we are conditioning on all of the
5 information in the set $\mathcal{F}$.

6 A typical application of this idea is when

7
$$\mathcal{F}_t = \text{information available at time } t$$

8 Then, we have that

9
$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$$

10 This sequence $\{\mathcal{F}_t\}$ is called a **filtration**.

11 For example, any trading strategy implemented at time t should condition
12 on $\mathcal{F}_t$, but not on $\mathcal{F}_{t+1}$.

Formally, such collections $\mathcal{F}$ are a σ -field or σ -algebra. A σ -**field** is a collection of subsets of Ω that satisfies three properties:

1. $\emptyset \in \mathcal{F}$
2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$
3. If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_i A_i \in \mathcal{F}$

Exercise: Comment on how each of the three properties of a σ -field relate to the interpretation that $\mathcal{F}$ represents information.

① I always know $\emptyset$ has not occurred

② If I know the occ/non-occ of A , then also know occ/non-occ of A^c

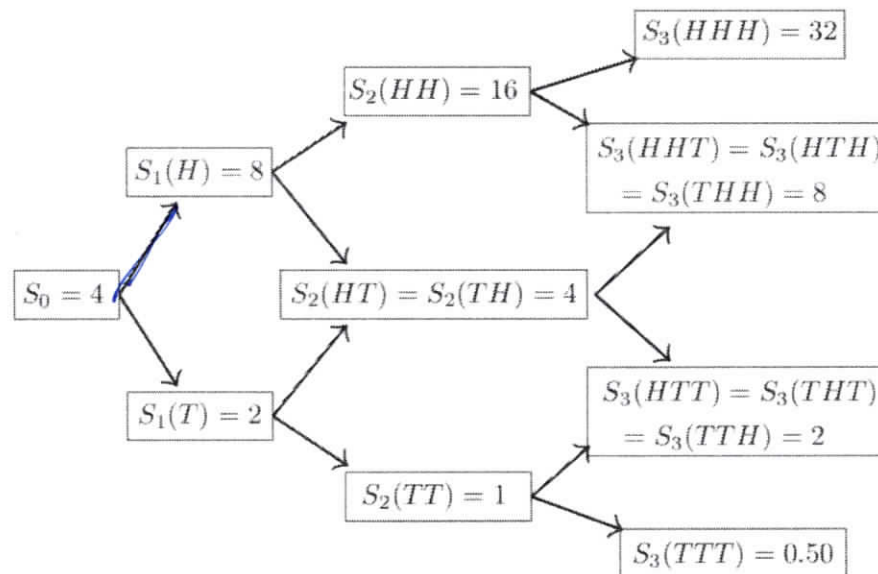
③ If I know the occ/non-occ of $A_1, A_2, \dots$, then I know the occ/non-occ of $\bigcup_{i=1}^{\infty} A_i$

Comments

1. Be careful with the notation and the interpretation: If $\mathcal{F}$ is a σ -field, its elements are subsets of Ω , i.e., if $A \in \mathcal{F}$, then $A \subseteq \Omega$.
2. The **trivial σ -field** consists of only $\emptyset$ and Ω . So, if $\mathcal{F}_0 = \{\emptyset, \Omega\}$, then conditioning on $\mathcal{F}_0$ is like conditioning on no information.
3. The **power set**, denoted 2^Ω , is a σ -field. It corresponds to “knowing everything.”
4. The information content from conditioning on a random variable Y is called the **σ -field generated by Y** , and is denoted $\sigma(Y)$. So, in our initial example, we were listing the sets in $\sigma(X)$. When we write $E(Y|X)$, that can be interpreted as $E(Y|\sigma(X))$.
5. A random variable X is said to be **$\mathcal{F}$ -measurable** if $\sigma(X) \subseteq \mathcal{F}$. Heuristically, this means that the information on which I am conditioning includes the value of X .

$$\text{e.g. } \sigma_{\mathcal{F}} = \sigma(X, Y) \\ \sigma(X) \subseteq \sigma(X, Y)$$

- Exercise:** Consider a simple three-period binomial pricing model. At each step, the price doubles with probability 0.5, and is cut in half with probability 0.5. All steps are independent.
- Start by constructing an appropriate Ω for this situation.
- Let S_t denote the value at time t , and assume that $S_0 = 4$.
- Find $E(S_t)$ for all values of t . What is $E(S_2 | \mathcal{F}_1)$? What is $E(S_3 | \mathcal{F}_1)$?



$$\Omega = \{HHH, HHT, \dots, TTT\}$$

notation $\omega \in \Omega$ write $\omega = \omega_1 \omega_2 \omega_3$
 $\omega_i = H \text{ or } T$

$$S_0(\omega) = 4 \quad \text{for all } \omega \in \Omega$$

$$S_1(\omega) = \begin{cases} 8, & \omega_1 = H \\ 2, & \omega_1 = T \end{cases}$$

$$E(S_0) = 4, \quad E(S_1) = 5, \quad E(S_2) = \frac{25}{4}$$

$$S_2(\omega) = \begin{cases} 16, & \omega_1 \omega_2 = HH \\ 4, & \omega_1 \omega_2 = HT \text{ or } \omega_1 \omega_2 = TH \\ 1, & \omega_1 \omega_2 = TT \end{cases}$$

What is $E(S_2 | \mathcal{F}_1)$? This is a RV,
it is a function on Ω .

If I know $\omega_1 = H$, then 50% chance
that $S_2 = 16$ and 50% chance that
 $S_2 = 4$. Gives me an EV of 10.

If I know $\omega_1 = T$, then 50% chance
that $S_2 = 4$, 50% chance that $S_2 = 1$
Gives me an EV of $5/2$.

S_0

$$E(S_2 | \mathcal{F}_1)(\omega) = \begin{cases} 10, & \omega \in \{HHH, HHT, HTH, HTT\} \\ 5/2, & \omega \in \{THH, TTH, THT, TTT\} \end{cases}$$

Note that $\mathcal{F}_2$ contains $\emptyset, \Omega, \{HHH, HHT, HTH, HTT\}, \{THH, TTH, THT, TTT\}$ and that's it

Here,

$$\sigma(S_1) = \mathcal{F}_1$$

Does it hold that

$$E(E(S_2 | \mathcal{F}_1)) = E(S_2) ?$$

Yes, " "

$$(10)(0.5) + \left(\frac{5}{2}\right)(0.5) = \frac{25}{4} = E(S_2)$$

$$E(S_3 | \mathcal{F}_1)(\omega) = (32)\left(\frac{1}{4}\right) + (8)\left(\frac{1}{2}\right) + (2)\left(\frac{1}{4}\right) \\ = 12.5$$

when $\omega_1 = H$, i.e. $\omega \in \{HHH, HHT, HTH, HTT\}$

$$E(S_3 | \mathcal{F}_1)(\omega) = (8)\left(\frac{1}{4}\right) + (2)\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) \\ = \frac{25}{8}$$

when $\omega_1 = T$

$E(S_3 | \mathcal{F}_1)$ is a RV

$$E(E(S_3 | \mathcal{F}_1)) = (12.5)(0.5) + \left(\frac{25}{8}\right)(0.5) \\ = E(S_3)$$

1 Properties of Conditional Expectation

Assume that $\mathcal{F}$ and $\mathcal{G}$ are σ -fields, and that X and Y are random variables.

$$1. E(E(X|\mathcal{F})|\mathcal{G}) = E(X|\mathcal{F}) \text{ if } \mathcal{F} \subseteq \mathcal{G}$$

$$2. E(E(X|\mathcal{G})|\mathcal{F}) = E(X|\mathcal{F}) \text{ if } \mathcal{F} \subseteq \mathcal{G}$$

$$3. E(E(X|\mathcal{G})) = E(X) \quad \leftarrow \mathcal{F} = \{\emptyset, \Omega\}$$

$$4. \text{ If } X \text{ is } \mathcal{F}\text{-measurable, then } E(X|\mathcal{F}) = X$$

$$\sigma(X) \subseteq \mathcal{F} \quad (\text{e.g. } \mathcal{F}_1 \subseteq \mathcal{F}_2)$$

$$5. \text{ If } X \text{ is } \mathcal{F}\text{-measurable, then } E(XY|\mathcal{F}) = XE(Y|\mathcal{F})$$

$$6. E(X + Y|\mathcal{F}) = E(X|\mathcal{F}) + E(Y|\mathcal{F}) \text{ (under technical conditions)}$$

$$7. E(cX + d|\mathcal{F}) = cE(X|\mathcal{F}) + d \text{ for scalars } c \text{ and } d$$

$$8. \text{ If } X \leq Y, \text{ then } E(X|\mathcal{F}) \leq E(Y|\mathcal{F})$$

$$9. \text{ If } \phi \text{ is convex, then } E(\phi(X)|\mathcal{F}) \geq \phi(E(X|\mathcal{F}))$$

A Final Definition: We say that $\mathcal{F}$ and $\mathcal{G}$ are **independent** if for any $A \in \mathcal{F}$ and $B \in \mathcal{G}$, it holds that $P(A \cap B) = P(A)P(B)$.

So, when we say that X and Y are independent, from a measure-theoretic standpoint we mean that $\sigma(X)$ and $\sigma(Y)$ are independent.

Part 5: Moment Generating Functions

Suppose random variable X has some univariate distribution, discrete or continuous.

We define $E(X)$, $E(X^2)$, $E(X^3)$, $\dots$ to be the **moments of the distribution of X** . As we've already seen, $E(X^k)$ can be finite, infinite, or not exist.

The **moment generating function (MGF)** of X is

$$\underline{M_X(t)} = E(e^{tX}).$$

Note that it is a function of t .

The MGF is simply the expected value of some function of X , and hence can be found using the Rule for the Lazy Statistician.

Exercise: Derive the MGF for the Poisson(λ) distribution.

$$X \sim \text{Poisson}(\lambda)$$

$$M_X(t) = E(e^{tx})$$

$$= \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda} \lambda^k}{k!}$$

Rule for
Lazy Stat.

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!}$$

$$= e^{-\lambda} e^{\lambda e^t}$$
$$= e^{\lambda(e^t - 1)}$$

$$e^a = \sum_{i=0}^{\infty} \frac{a^i}{i!}$$

Exercise: Derive the MGF for the Exponential(λ) distribution.

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

Name	Density/Mass Function	MGF	Mean	Variance
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$ $-\infty < x < \infty, \sigma > 0$	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$	μ	σ^2
Log-Normal	$\frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(x)-\mu)^2}{2\sigma^2}\right),$ $0 < x < \infty, \sigma > 0$	—	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$	$e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}$
Gamma	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x},$ $0 < x < \infty, \alpha, \beta > 0$	$\left(\frac{\beta}{\beta-t}\right)^\alpha, t < \beta$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Exponential	$\beta e^{-\beta x},$ $0 < x < \infty, \beta > 0$	$\frac{\beta}{\beta-t}, t < \beta$	$\frac{1}{\beta}$	$\frac{1}{\beta^2}$
Beta	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1},$ $0 < x < 1, \alpha, \beta > 0$	$1 + \sum_{k=1}^{\infty} \left(\prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r}\right) \frac{t^k}{k!}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Uniform (continuous)	$\frac{1}{b-a},$ $a \leq x \leq b$	$\frac{e^{bt}-e^{at}}{t(b-a)}, t \neq 0$ $1, t = 0$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Bernoulli	$p^x(1-p)^{1-x},$ $x = 0, 1, 0 \leq p \leq 1$	$pe^t + (1-p)$	p	$p(1-p)$
Binomial	$\binom{n}{x} p^x(1-p)^{n-x},$ $x = 0, 1, \dots, n, 0 \leq p \leq 1$	$[pe^t + (1-p)]^n$	np	$np(1-p)$
Negative Binomial	$\binom{r+x-1}{x} p^r(1-p)^x,$ $x = 0, 1, \dots, r \in \{1, 2, \dots\}, 0 \leq p \leq 1$	$\left(\frac{p}{1-(1-p)e^t}\right)^r, t < \log(1/(1-p))$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Geometric	$(1-p)^x p,$ $x = 0, 1, \dots, 0 \leq p \leq 1$	$\frac{p}{1-(1-p)e^t}, t < \log(1/(1-p))$	$\frac{1-p}{p}$	$\frac{(1-p)}{p^2}$
Poisson	$\frac{e^{-\lambda} \lambda^x}{x!},$ $x = 0, 1, \dots, \lambda > 0$	$e^{\lambda(e^t-1)}$	λ	λ
Hypergeometric	$\frac{\binom{A}{x} \binom{B}{n-x}}{\binom{A+B}{n}},$ $x = 0, 1, \dots, n$	—	$\frac{nA}{A+B}$	$\frac{nAB(A+B-n)}{(A+B)^2(A+B+1)}$

Using the MGF

1. Calculating Moments

Successive derivatives of the MGF, evaluated at $t = 0$, return the moments of the random variable:

$$E(X^k) = \left. \frac{\partial^k}{\partial t^k} M_X(t) \right|_{t=0}$$

Exercise: Why does this work? Sketch it out in the continuous case.

$$\frac{\partial^k}{\partial t^k} M_X(t) = \frac{\partial^k}{\partial t^k} \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$$

$$= \int_{-\infty}^{\infty} \frac{\partial^k}{\partial t^k} e^{tx} f_X(x) dx$$

$$= \int_{-\infty}^{\infty} x^k e^{tx} f_X(x) dx$$

plus in $t=0$

$$= \int_{-\infty}^{\infty} x^k f_X(x) dx = E(X^k)$$

Exercise: Find the moments of the Poisson(λ) distribution.

$$X \sim \text{Poisson}(\lambda)$$

$$M_X(t) = e^{\lambda(e^t - 1)}$$

$$\frac{d}{dt} M_X(t) = e^{\lambda(e^t - 1)} \lambda e^t$$

$$\text{plus in } t=0, E(X) = \lambda$$

$$\frac{d^2}{dt^2} M_X(t) = \frac{d}{dt} e^{\lambda(e^t - 1)} \lambda e^t$$

$$\text{plus in } t=0, E(X^2) = \lambda^2 + \lambda$$

2. Uniqueness of the MGF

The uniqueness result is the following: If $M_X(t) = M_Y(t)$ for all t , then X and Y have the same distribution.

It is interesting to note that two random variables can have matching moments, i.e., $E(X^k) = E(Y^k)$ for $k = 1, 2, \dots$, but X and Y have different distributions.

Exercise: Suppose that X has the $\text{Exponential}(\lambda)$ distribution, and $k > 0$ is some constant. Does kX have an Exponential distribution?

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

$$\text{Let } Y = kX$$

$$M_Y(t) = E(e^{tY}) = E(e^{tkX})$$

$$= M_X(tk) = \frac{\lambda}{\lambda - tk} = \frac{\lambda/k}{\lambda/k - t}$$

matches MGF for $\text{Exponential}(\lambda/k)$

So, $Y \sim \text{Exponential}(\lambda/k)$

1 **Exercise:** Suppose that the number of events during an interval of
 2 length one time unit is a random variable X which has the $\text{Poisson}(\lambda)$
 3 distribution. For each of these events, there is probability p that it is
 4 of "Type I." The determination of whether or not an event is of "Type
 5 I" is independent from event to event. Let Y denote the number
 6 of "Type I" events during the time interval. Show that Y has the
 7 Poisson distribution.

8 we know $X \sim \text{Poisson}(\lambda)$, $Y|X \sim \text{binomial}(X, p)$

9 $M_Y(t) = E(e^{tY})$

11 $= E(E(e^{tY}|X))$ using iterated conditioning

13 $= E((pe^t + (1-p))^X)$ using $Y|X \sim \text{binom}$

15 $= \sum_{k=0}^{\infty} (pe^t + (1-p))^k \frac{e^{-\lambda} \lambda^k}{k!}$

17 $= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda(pe^t + (1-p)))^k}{k!}$

19 $= e^{-\lambda} e^{\lambda(pe^t + (1-p))}$

21 $= e^{\lambda p(e^t - 1)}$ matches $\text{Poisson}(\lambda p)$ MGF

22 so, $Y \sim \text{Poisson}(\lambda p)$

3. Sums of Independent Random Variables

Suppose that $X_1, X_2, \dots, X_n$ are independent random variables, and

let $Y = \sum_i X_i$. Then,

$$M_Y(t) = \prod_{i=1}^n M_{X_i}(t)$$

for all t such that all $M_{X_i}(t)$ is finite.

Exercise: Prove the above result.

$$M_Y(t) = E(e^{tY})$$

$$= E(e^{t \sum X_i})$$

$$= E(e^{tX_1} e^{tX_2} \dots e^{tX_n})$$

$$= E(e^{tX_1}) E(e^{tX_2}) \dots E(e^{tX_n})$$

$$= \prod_{i=1}^n M_{X_i}(t)$$

Exercise: Show that the sum of independent Poisson random variables also has the Poisson distribution.

$$X_i \sim \text{Poisson}(\lambda_i) \quad i=1,2 \quad \text{ind.}$$

$$Y = X_1 + X_2$$

$$M_Y(t) = M_{X_1}(t) M_{X_2}(t) = e^{(\lambda_1 + \lambda_2)(e^t - 1)}$$

This is the MGF for $\text{Poisson}(\lambda_1 + \lambda_2)$

Hence, $Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$

Exercise: Under what condition(s) does the sum of Gamma random variables still have a Gamma distribution?

$$X_i \sim \text{Gamma}(\alpha_i, \beta_i) \quad i=1,2,\dots,n$$

$$M_{X_i}(t) = \left(\frac{\beta_i}{\beta_i - t} \right)^{\alpha_i}, \quad t < \beta_i$$

Assume $X_1, X_2, \dots, X_n$ independent. Let $Y = \sum X_i$

$$M_Y(t) = \prod_{i=1}^n \left(\frac{\beta_i}{\beta_i - t} \right)^{\alpha_i}$$

In order for this to remain Gamma need $\beta_1 = \beta_2 = \dots = \beta_n = \beta$. Then

$$M_Y(t) = \left(\frac{\beta}{\beta - t} \right)^{\sum \alpha_i}$$

which is $\text{Gamma}(\sum \alpha_i, \beta)$

Exercise: Prove the following properties of the Gamma distribution.

1. If $X_1, X_2, \dots, X_n$ are iid $\text{Exponential}(\lambda)$ random variables, the $\sum_i X_i$ is $\text{Gamma}(n, \lambda)$.
2. If X has the $\text{Gamma}(\alpha, \beta)$ distribution and $k > 0$, then kX is $\text{Gamma}(\alpha, \beta/k)$.

4. Establishing Convergence in Distribution

In some situations it will be important to formalize the notion of “the distribution of $X_1, X_2, \dots$ converges to the distribution of X .”

In such cases, we would write $X_n \xrightarrow{D} X$, and say that “ X_n converges in distribution to X .”

The formal definition is as follows: X_n **converges in distribution** to X if $F_{X_n}(x) \rightarrow F_X(x)$ as $n \rightarrow \infty$ at all x where F_X is continuous.

In practical terms, the value of knowing $X_n \xrightarrow{D} X$ is that (for large n), the distribution of X can be taken as an approximation to the distribution of X_n . The **central limit theorem** is an important example.

The useful result: $X_n \xrightarrow{D} X$ if $M_{X_n}(t) \rightarrow M_X(t)$ as $n \rightarrow \infty$ for all $-\epsilon < t < \epsilon$ for some $\epsilon > 0$.

1 **Exercise:** Suppose that X equals zero with probability one and X_n
2 equals $1/n$ with probability one. Does $X_n \xrightarrow{D} X$?

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1 **Exercise:** Suppose that X_n has the Binomial($n, \lambda/n$) distribution for
2 $\lambda > 0$. Let X have the Poisson(λ) distribution. Show that $X_n \xrightarrow{D} X$.

$$\begin{aligned} 3 \quad M_{X_n}(t) &= \left(1 - \frac{\lambda}{n} + \frac{\lambda}{n} e^t\right)^n \\ 4 \quad &= \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \end{aligned}$$

$$7 \quad \lim_{n \rightarrow \infty} M_{X_n}(t) = \lim_{n \rightarrow \infty} \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n$$

$$10 \quad = e^{\lambda(e^t - 1)}$$

$$12 \quad \boxed{e^a = \lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n}$$

$$15 \quad = M_X(t)$$

$$17 \quad \text{So, } X_n \xrightarrow{D} X$$